

Quiz 2

1. C/D definition

Write the mathematical relationship between a continuous-time signal $x_c(t)$ and its samples $x[n]$. Define T and f_s .

2. Sampled Spectrum

For ideal impulse-train sampling, write $X_s(j\Omega)$ in terms of $X_c(j\Omega)$, T , and Ω_s . What physical picture does this equation describe?

3. Alias calculation

A 9 kHz sinusoid is sampled at 12 kHz. What sinusoidal frequency appears in the sampled data when interpreted in $[0,6]$ kHz?

4. Nyquist condition

A signal is strictly bandlimited to 4.5 kHz. What is its ideal Nyquist rate? Is $f_s = 8$ kHz sufficient?

5. D/A concept

What frequency-domain magnitude effect is introduced by a zero-order hold?

6. Quantization

Estimate the ideal full-scale sine SQNR for a 10-bit uniform quantizer.